

Code-Layout, Readability and Reusability

Date	28 June 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	PEP-8 indentation standards followed
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for templates, static files, and model files
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as life_expectancy, education_index, predicted_hdi are used
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as predict_hdi() and load_model() are used
5	Code Comments	Inline and block comments explain logic	Yes	Important sections are documented with comments
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate modules for preprocessing, prediction, and visualization
7	No Redundant Code	Duplicate or unused code is removed	Yes	Code is optimized and duplicate code is eliminated
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic input validation and exception handling are implemented

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python, Pandas	Dataset cleaning and preprocessing	High

2	HDI Prediction Model	Python, Scikit-learn	Prediction functionality in Flask application	High
3	Visualization Module	Matplotlib, Seaborn	Generating charts and reports	High
4	Flask Backend Module	Python, Flask	Handling user requests and predictions	Medium
5	HTML Templates	HTML, CSS	User interface pages	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-structured folder hierarchy and organized files
Readability	5	Meaningful names and proper formatting improve readability
Reusability	4	Modules can be reused in future enhancements
Documentation / Comments	4	Adequate comments are provided for important code sections
Overall Score	4.5	The codebase is maintainable, reusable, and easy to understand